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# PIR - 61 wine products trading at incorrect prices across Scotland, Ireland & Wales stores.

## M&S Digital&Tech *Post Incident Review*



**Incident/ Problem Reference: 90661983**

**Incident Date: 14/03/2025**

**PIR Date: 18/03/2025**

**Author: Saloni Chowdhury**

**Technical Owner: Jaturson Yogendran/ Muthuvelan Maheswaran**

**PIR Reviewer: Samantha Welsh /Thomas Grigg**

### 1. Incident Summary :

On 14/03, 61 wine products across 116 stores in Scotland, Ireland and Wales were scanning at incorrect prices on store tills between 10:15 and 12:45. Approx. 500 customers were impacted resulting in poor customer experience and a finance loss of £5.5k. The root cause was attributed to a human error whilst updating the unit quantity for multiple wine products as requested by the Business. Services were restored after the prices were manually corrected in POS and sent to the store tills.

### 2. Incident Chronology :

12:57 Wales stores have confirmed resolution

12:50 4 stores from Scotland have confirmed resolution, awaiting confirmation from Wales and Ireland

12:42 The fix is deployed successfully, and teams are validating with stores

12:34 - ETA 7 mins for tills to receive the updated prices

12:30 script is deployed - which will update in CSSM and then send to Tills. Team will validate the changes in CSSM.

12:00: POS team are currently creating an XML to revert the price changes to the old price across 61 impacted UPCs.

11:57: From POS, we have identified 61 impacted UPCs with reduced and increased prices. In SAP Master Data, we have observed min quantity (ml to cl) changes across 354 UPCs on 22nd Jan, as per the request from business to perform weight changes. On 13th March, the changes were reverted from cl to ml.

11:44 : Galileo team to share the list of UPCs with change in quantity/prices. POS team to first revert the quantity changes followed by price changes.

11:32 2 options are being discussed - New price with min unit price / end the incorrect price

11:23 We have identified ~350 UPCs which are potentially impacted.

11:19 Ireland seems to be part of the impacted list - impacted stores to be collected

11:17 Stores communication have been sent out to avoid further impacts - ETA 1 hour to get back to business with a workaround

11:15 POS team can manually update the prices( as a workaround ) , so that customers can get the right prices ( POS overnight prices

11:07 We have identified a change made to the net content for a sample article in SAP Master data system by the Galileo team - which means that

the selling price which is calculated based on the quantity of the articles. It is understood, that the change in net content has overridden the selling price of the articles.

| Article 60628287<br>Changed by P9197240 Date 13.03.2025 Time 13:39:30 |           |                       |            |            |            |
|-----------------------------------------------------------------------|-----------|-----------------------|------------|------------|------------|
| Promo                                                                 | Org. Unit | Field Descript.       | Old Value  | New Value  | Additional |
| Change                                                                |           | Net contents          | 75.000 CL  | 750.000 CL |            |
| Change                                                                |           | Date of Last Change   | 03.03.2025 | 13.03.2025 |            |
| Change                                                                |           | Net Weight            | 75.000 G   | 750.000 G  |            |
| Change                                                                |           | Comparison price unit | 75         | 100        |            |

11:03 Selling price seems to be changed by business for 1 UPC, SAP is validating the changes

11:02 Technical teams are on a call

10:10 Service Lead was reported that Multiple Wine UPC's scanning in incorrect Price in Tills in Wales and Scotland stores

### 3. Incident Resolution

An analysis was conducted to identify the impacted wine products with incorrect prices. The prices were then manually corrected in POS to reflect the previous prices and sent to the store tills.

Additionally, the net content of the 61 impacted UPCs were corrected within SAP Master Data to prevent further issues with prices at store tills after overnight processing.

### 4. Root Cause

The root cause was attributed to a human error whilst reverting the changes on 61 out of 354 wine products by the Galileo team where the unit of measure (cl to ml) and the net content was incorrectly updated in SAP master data system.

#### Context:

On 22nd January, the Galileo Trading Team received a request from Food Catalog Team to:

1. Update the content unit from milliliter (ML) to centiliter (CL).
2. Update the comparison price unit to 75 CL across 354 alcohol lines (no start date specified).

The change aimed to display a per 75 CL value on the online food catalog. Since updating ML to CL was unfamiliar, the Galileo team proposed an initial test on five products before applying it across all items.

On 23rd January, Galileo team tested updates on five lines across two SAP interfaces (BASIC & View Article by Date - VABD). This was validated by business and therefore, updates for the full list were approved.

On 28th January, the net content updates were applied to 340/354 products successfully. However, 14 products faced a decimal value issue (37.5 CL → 375 ML) in SAP VABD screen. SAP support was engaged but did not provide a resolution until 5th March, confirming that a technical configuration update was required.

On 12th March, business (Foods trading Manager & Foods catalogue team) requested a full reversion of the changes which was performed by the Galileo team on 13/03 before the SAP cut-off. This activity was split across multiple team members due to other priorities within the team.

The end-to-end flow is as follows: **SAP → Enterprise Services → FMD → DS → POS.**

The Galileo team identified the issue during validation steps with SAP with 61 products. However, due to the nature of the changes being made, it was initially assumed that there would be no significant impact on downstream systems. In parallel, stores raised issues, triggering an incident bridge. To prevent similar issues in the future, we have agreed on an action to perform validations on the same day a change is made in SAP. This will involve extracting a report from SAP to ensure the changes have been implemented correctly.

Additionally, the issue did not impact ECS tickets or any other downstream systems, as minimum selling price is only used by POS, whereas ECS tickets rely on the standard retail price.

**Recovery options:** During the incident call, there were discussions about an option to edit the LALO (Lowest Available Allowable) price in POS to prevent illegal trading scenarios by allowing store colleagues to edit the prices directly in POS. However, POS currently does not have this functionality. As a result, we have agreed on an action to explore the feasibility of a configuration change that can be enabled during an illegal trading scenario to allow temporary price adjustments. Once the issue is resolved, the configuration can be disabled to prevent any manual price overrides by store colleagues.

#### Process Improvements:

To enhance the changes made by the Business Process Services team in response to business requests and to prevent future incidents, an exercise should be conducted to identify SAP master data fields that trigger the recalculation of the minimum selling price across the UK and other countries.

So far, we have identified the following fields that impact minimum selling price recalculation:

- Net content
- Content unit
- Alcohol percentage

Manual errors whilst making changes to these fields can have a cascading effect on minimum prices in POS, potentially leading to pricing issues.

Once all relevant fields are identified, the Business Process Services team should follow the change process. This will ensure that post-implementation checks are properly aligned to detect and resolve issues promptly, minimizing disruption for store colleagues.

## 5. Findings

### 5.1. Observability –

The issue was highlighted by stores involving wine products scanning at tills with incorrect prices due to changes in net content requested by the business. Since this was a manual activity, we concluded that introducing an alert mechanism to detect errors in these changes was not feasible. However, to mitigate potential discrepancies, we have agreed to extract a report after any modifications are made in the SAP master data, allowing us to validate the changes before they are sent to downstream systems.

### 5.2. Technical Challenges -

There were no technical challenges in recovering the incident. However, during the PIR, we discussed implementing a systemic validation step in SAP to automatically verify changes in master data without manual intervention. It was ultimately concluded that the existing SAP code does not support systemic restrictions, so for now, the validation process will remain manual.

### 5.3. MIM involvement & impact assessment -

Although the MIM was engaged promptly and the impact on 61 products was identified quickly, the Galileo team, when validating the changes with SAP, discovered incorrect modifications in the master data for these products. However, they initially assumed that the quantity change would not affect the minimum prices sent to the tills. To prevent similar issues, we have agreed on an action to identify the master data fields that, when changed, can impact minimum selling price calculations. So far, we have identified **net content, content unit, and alcohol percentage** as key fields that trigger recalculations. Additionally, to improve the process followed by the Business Process Services team, we are working on establishing a change management framework for modifications to these critical master data fields.

### 5.4. Incident handling/ stakeholder management –

There were no concerns in managing the incident, as all relevant stakeholders, including representatives from the POS, Galileo, and SAP teams, were actively involved in the call and carried out the recovery actions promptly.

### 5.5. Business Communication –

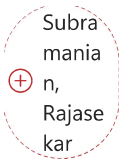
The service lead sent business communications in a timely manner; however, feedback was received suggesting that the retail impact on stores should be more clearly detailed for better visibility among retail business stakeholders.

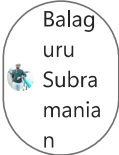
Additionally, we have agreed to revisit the incident's categorization, while it was classified as a **significant** incident based on financial loss, no. of stores under the impact framework, feedback was received suggesting it should be categorized as **Major** due to the illegal trading scenario. The service lead will further engage with stakeholders to clarify the categorization as we advised stores to stop selling the affected wine products to prevent illegal trading..

### 5.6. Supplier Engagement –

There were no supplier engagement in this incident.

## Mitigation Actions:

|   | ≡ Action                                                                                                                                                                                                                                                   | ≡ Owner                                                                                                       | ≡ Due Date | ≡ Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 📌 Action Category   |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| 1 | Standard Operating Procedure (SOP) document will be created by 31st March for this new request, including highlighting all necessary checks and corresponding updates [content unit update (unit a -> unit b) and net content value (value a -> value b)]. |  Yogen dran, Jaturs on       | 31st March | The SOP creation is in progress.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Problem Avoidance   |
| 2 | Additional associate training / best practice sessions will be conducted by the Subject Matter Expert to ensure all associates are familiar with executing this update                                                                                     |  Yogen dran, Jaturs on       | 31st March | In progress                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Solution & Recovery |
| 3 | Introduce a peer review process for master data changes to ensure their accuracy before being sent to downstream systems, particularly POS.                                                                                                                |  Aravin d, Lilly             | Completed  | <p>Once changes are made in SAP master data, a report will be extracted on the same day before the SAP cutoff at 16:30 to validate the modifications. Any anomalies identified will be corrected in collaboration with the SAP team before the cut off.</p> <p>If discrepancies are detected after the data has been sent to downstream systems, reactive recovery will be initiated through the major incident management process, involving POS, FMD, and other downstream systems to implement manual corrections. POS has an existing capability to adjust prices before stores opens.</p> | Solution & Recovery |
| 4 | Explore option to validate the changes in SAP within Enterprise Services for additional checks before the overnight batch kicks in.                                                                                                                        |  Subra mania n, Rajase kar | 31st March | The feasibility is being explored.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Solution & Recovery |
|   |                                                                                                                                                                                                                                                            |                                                                                                               |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                     |

|   |                                                                                                                                                                                                                                     |                                                                                                                                                                       |               |                                                                                                                                                                                                                                                                                                                                                                         |                   |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| 5 | Explore options for allowing store colleagues to edit the Lowest Available Allowable (LALO) price on the tills during an illegal trading scenario to prevent trading impact.                                                        | <br> | 31st March    | Teams will assess the feasibility of creating a central configuration that enables stores to temporarily adjust prices in such scenarios. Once the issue is resolved, this configuration will be disabled to prevent any unauthorised price overrides in POS.                                                                                                           | Problem Avoidance |
| 6 | Align on the impact framework to streamline incident categorization, considering that the illegal trading scenario was mitigated by advising stores to stop selling the affected wine products                                      |                                                                                      | 25th March    | Valli will discuss with Rich to determine the appropriate categorization of incidents involving illegal trading scenarios.                                                                                                                                                                                                                                              | Impact assessment |
| 7 | Explore the feasibility of configuring a technical or quality-of-life change within SAP that either couples or auto-converts unit of measure updates or provides a notification when a corresponding change is not made accurately. |                                                                                      | Not feasible. | During the PIR, the SAP architect confirmed that deploying systemic restrictions within the SAP standard code is not feasible. As a result, validation will remain a manual process. To mitigate risks, we have agreed on the actions outlined above to ensure changes are made accurately and that any errors can be identified and corrected before impacting stores. | Problem Avoidance |

Please refer to  Major\_Significant and Key Incident Tracker v1.0.xlsx for progress updates on Mitigations

<Mandatory participants: Put it in the meeting invite. >

## Technical Details and Useful Documents: